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Catching Up with the Fastest Star in the Galaxy: A Two-Passage GRAVITY+ Campaign to Detect S301's Schwarzschild Precession

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ABSTRACT

S301 is the fastest known star in the Galaxy. It orbits Sgr A* every 8.68 yr on an $e = 0.9832$ orbit that carries it within 136 Schwarzschild radii at 25 600 km s^{-1} (GRAVITY Collaboration et al. 2026). An orbit like that must precess, by the same Schwarzschild effect that turns Mercury's perihelion, whatever the spin of the black hole. We ask whether an astrometry-only GRAVITY+ campaign covering the 2031.8 and 2040.5 periapsis passages can catch it. We validate a relativistic light-curve model, design a 131-epoch Fisher-optimal cadence with a periapsis floor, fit a seven-parameter orbit including the light-travel-time delay, and build a systematic error budget. With white noise at the 207 μas achieved on S301, the fit recovers the injected rate, 0.2307 deg yr^{-1}, as 0.2324 ± 0.0023 deg yr^{-1} (± 0.0012 at the forecast 100 μas). Real astrometric noise is not white. We add per-run calibration offsets and a wandering Sgr A* photocentre at GRAVITY's published amplitudes, marginalize over the reference zero point, mass and distance, add the bounded systematics in quadrature, and inflate by the excess scatter of real GRAVITY orbit fits. What survives is 0.0074 deg yr^{-1}, a 31σ detection. The Sun hides the first periapsis for 153 d, at a cost to the period and periapsis epoch but not to the precession rate. Radial velocities are out of reach at S301's magnitude; photometry only validates the light-curve model. The spin of Sgr A* could shift the rate by 5.17–5.74 per cent and cannot be bounded, so the campaign is a spin-agnostic test of general relativity, and a forecast until the star comes round.

Keywords: Galactic Center — Supermassive black holes — Astrometry — General relativity — Orbit determination — Observational astronomy

1. INTRODUCTION

The S-stars around Sgr A* have delivered the strongest tests of general relativity in the field of a supermassive black hole: the gravitational redshift and the Schwarzschild precession of S2 (GRAVITY Collaboration et al. 2018a, 2020). S301 (GRAVITY Collaboration et al. 2026) is a sharper probe: half the period of S2 (8.68 against ∼16 yr), a higher eccentricity, and a periapsis ten times closer (136 against ∼1400 Schwarzschild radii, R_S). It was found in a search for a star that could measure the spin of Sgr A* through Lense–Thirring precession; a companion paper (T. Piran et al. 2026) develops the multi-star method that measurement requires.

This study began with a simpler question. Picture the star at closest approach, 136 R_S from the horizon and moving at 8 per cent of the speed of light. Some of the light it sends our way must pass so near the black hole that it is bent aside or swallowed. Would we see the star dim as it grazes the shadow? Section 3.1 answers: the black hole takes almost nothing, and the star does change brightness, but through beaming and gravitational redshift, and by too little to detect at nineteenth magnitude. The relativistic signal worth chasing is in the astrometry. Any Schwarzschild black hole, spinning or not, turns the long axis of an orbit around it; for S301 the predicted rate is 0.2307 deg yr^{-1}, an order of magnitude faster than for S2. Can a campaign that could be proposed today catch it? Three things stand in the way. S301 is faint ($m_K = 19.3$), so only interferometric astrometry can follow it and no current spectrograph can measure its velocity. The field is crowded. And its next periapsis, in October 2031, falls behind the Sun.

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Table 1. Adopted S301 parameters (GRAVITY Collaboration et al. 2026), Solution 1.

Parameter	Value	Uncertainty
Period P	8.680 yr	0.110 yr
Eccentricity e	0.9832	0.0010
Inclination i	124.09°	1.10°
Node Ω	73.8°	3.5°
Argument of periapsis ω	293.4°	2.2°
Periapsis epoch t_0	2023.126 yr	0.010 yr
Semi-major axis a	83.0 mas (687 AU)	···
Sgr A* mass $M_\bullet$	$4.297 \times 10^6\, M_\odot$	···
Distance R_0	8277 pc	···
K magnitude m_K	19.3	0.3

We answer by building the complete pipeline an observing proposal would need: the adopted orbit (Section 2); a validated light-curve model, the choice of instrument and observables, the cadence and the orbit fit (Section 3); the recovered precession and tests of the cadence (Section 4); the systematic error budget, where the feasibility case is decided (Section 5); and what the test does and does not claim (Section 6). Every number below is produced by the accompanying code and inserted by macro, so the prose cannot drift from the calculation.

2. THE S301 ORBIT

All ground-truth parameters come from GRAVITY Collaboration et al. (2026), Extended Data Table 2, Solution 1 (Table 1); a second, nearly degenerate solution is published but not used. The discovery fit already includes the Rømer (light-travel-time) delay and Schwarzschild precession, so these elements are defined in the same relativistic model we use. At periapsis S301 passes 136 R_S from Sgr A* at 25 600 km s^{-1}, more than 8 per cent of c, closer and faster than any other known star. For comparison, S2 reaches ≈7650 km s^{-1} at its own periapsis (GRAVITY Collaboration et al. 2018a); the fastest star previously claimed near Sgr A*, S4714, reaches $23\,900 \pm 8\,800$ km s^{-1} (F. Peißker et al. 2020); and the fastest hypervelocity star known, S5-HVS1, is leaving the Galaxy at 1755 ± 50 km s^{-1} (S. E. Koposov et al. 2020). S301 is the fastest known star in the Galaxy. Its effective temperature is unpublished; we adopt 7300 K from standard F1.5V tables, which affects only the light-curve amplitude. The mass and distance of Sgr A* are those of GRAVITY Collaboration et al. (2022): $M_\bullet = (4.297 \pm 0.012) \times 10^6\, M_\odot$ and $R_0 = 8277 \pm 9$ pc (statistical), with systematic uncertainties of ≈ 40,000 $M_\odot$ and ≈ 30 pc.

3. METHOD

3.1. *Relativistic light curve*

We model S301's K-band light curve from the orbit of Table 1. The model applies the exact special-relativistic Doppler factor and the Schwarzschild gravitational redshift $\sqrt{1 - R_S/r}$ to a 7300 K blackbody, boosts the intensity as I_ν/ν^3, and integrates over a top-hat bandpass covering GRAVITY's K-band range, 1.98–2.40 μm (GRAVITY Collaboration et al. 2017). Three effects are omitted on purpose: lensing by Sgr A*, treated as an astrometric systematic in Section 5.6; spin-dependent (Kerr) terms, which would require a value for the unmeasured spin; and the Rømer delay, which does not change a curve plotted against time from periapsis and enters the fit instead (Section 3.5). The model reproduces the published solution: $a = 686.7$ AU against 687.0, periapsis distance 136.0 R_S against 136, and periapsis speed 8.54 per cent of c against the published ">8%". This validates the engine before we ask it to predict anything new.

The predicted periapsis signature is an asymmetric beaming spike (Fig. 3). The star brightens by 0.075 mag on approach and fades to 0.040 mag below baseline on recession, when de-boosting and gravitational redshift act together; the peak-to-trough amplitude is 0.115 mag. The spectral slope in the band sets the amplitude: at 2.2 μm and 7300 K,

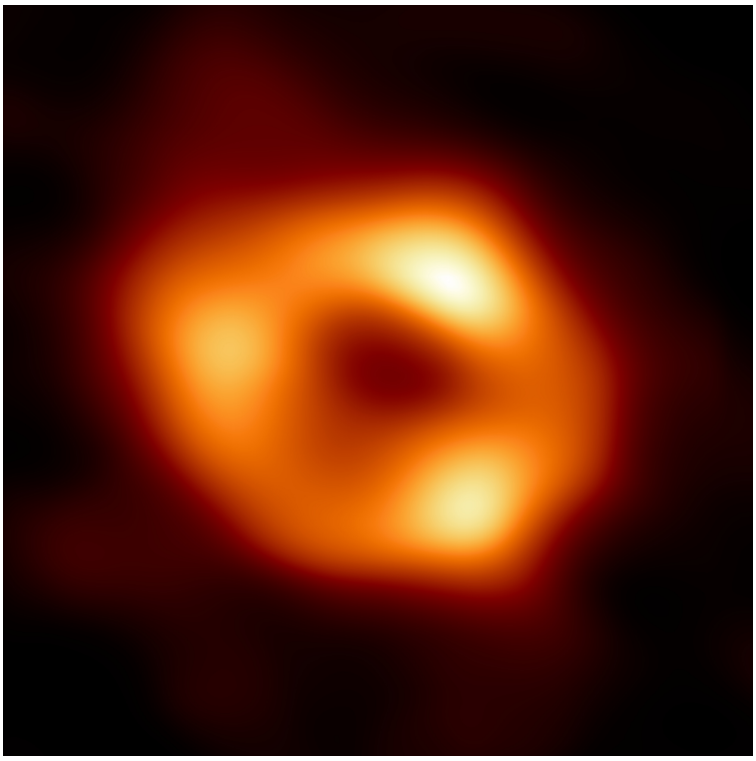


Figure 1. Sgr A* itself, for scale: the Event Horizon Telescope image of the black hole's shadow, ring diameter 51.8 ± 2.3 μas (Event Horizon Telescope Collaboration 2022), about 1600 times smaller than S301's 83 mas orbit and far below the astrometric resolution of this campaign. Image credit: EHT Collaboration, released by the European Southern Observatory (ESO) as eso2208-eht-mwa (2022-05-12), licensed CC BY 4.0.

$h\nu/kT \approx 0.9$, so the in-band flux scales roughly as $(1+z)^{-1.5}$, far more gently than the bolometric $(1+z)^{-4}$. Near periapsis the full relativistic redshift exceeds the classical Doppler shift by 0.07–0.72 percentage points.

The dimming deserves care, because it is not the dimming we first imagined. Sgr A* neither swallows nor deflects a measurable share of S301's light: at 136 R_S the horizon subtends a negligible fraction of the star's sky, and Section 5.6 shows that light bending amounts to an astrometric shift of at most 22.2 μas at a handful of epochs. The fading after periapsis is gravity acting on the light in a subtler way. Each photon loses energy climbing out of the potential well, on top of the Doppler de-boosting of a receding source; the black hole taxes the photons it lets through rather than taking them. There is no occultation to model, because Sgr A* is compact and blocks nothing.

3.2. *Instrument and observables*

GRAVITY+ is the upgraded beam combiner of the VLTI (Very Large Telescope Interferometer) that discovered S301. It combines the four 8.2 m Unit Telescopes at Paranal, each with adaptive optics, on baselines up to ∼130 m; the baseline, not the aperture, gives it sub-milliarcsecond astrometry in the crowded Galactic Centre. We adopt a per-epoch precision of 207 μas, the value achieved on S301 itself in the discovery data, where it is set by the pixel grid of the image reconstruction (GRAVITY Collaboration et al. 2026, Methods). The discovery paper's own forecast for the completed GRAVITY+ is 100 μas, "as expected for the final performance of GRAVITY+" (GRAVITY Collaboration et al. 2026); we treat that as the optimistic case and report it alongside (Section 4). Each epoch also yields K-band photometry, which we simulate with S301's published uncertainty of 0.30 mag but do not fit (Section 4.3).

Radial velocity was rejected before any data were simulated. SINFONI, the integral-field spectrograph formerly at the VLT (Very Large Telescope), reached 12.3 km s^{-1} on S2 at $m_K = 14.0$ (GRAVITY Collaboration et al. 2018a); its successor ERIS (Enhanced Resolution Imager and Spectrograph) carries the SPIFFIER spectrograph at resolving power $R = 5000$ (R. I. Davies & et al. 2023). S301 is 5.3 mag fainter than S2 (∼132× less flux), so in the background-limited regime the SINFONI benchmark scales to ∼1621 km s^{-1} per epoch: 0.11 of S301's ±15285 km s^{-1} velocity swing, and only 0.037 of SPIFFIER's 60 km s^{-1} resolution element. A line centroid cannot be located to so small a fraction of a resolution element at this flux, and the discovery team found the same at the telescope: "Our current

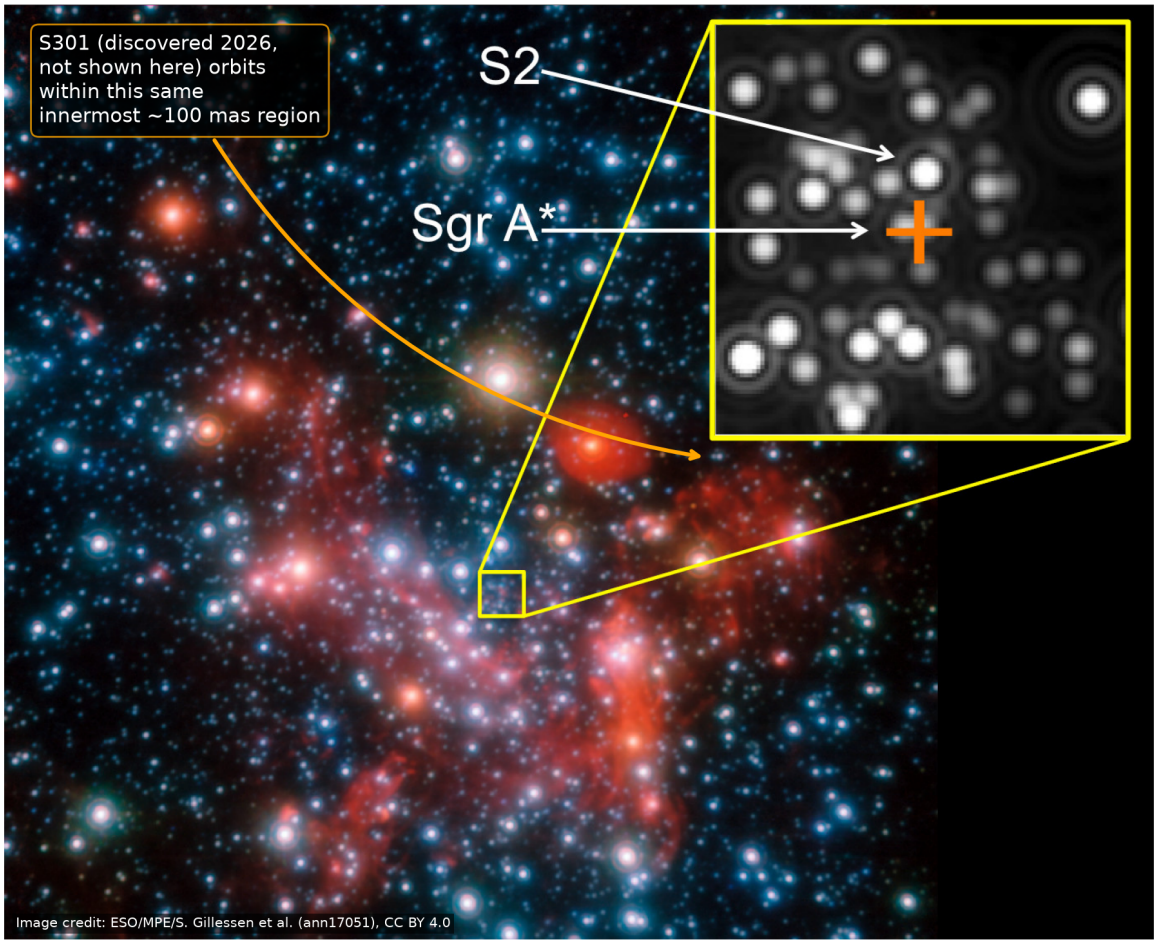


Figure 2. The crowded innermost Galactic Centre in the near-infrared (a real image), with the S2/Sgr A* zoom inset from the original release. S301 was discovered in 2026, after this 2017 image, and is not resolved in it; the schematic annotation shows that its 83 mas orbit lies in the same innermost few tens of mas, the crowding relevant to Section 5.5. Image credit: ESO/MPE/S. Gillessen et al., released as ann17051 (2017-08-09), licensed CC BY 4.0.

ERIS spectroscopy is not deep enough" (GRAVITY Collaboration et al. 2026). Section 6.2 confirms by simulation that an ERIS channel would not improve any parameter and assesses the ELT (Extremely Large Telescope) prospects.

3.3. *Cadence design*

An orbit this eccentric is mostly waiting. S301 drifts through apoapsis for the better part of a decade and then sweeps through most of its angular track in a few weeks around periapsis; a cadence that misses those weeks has watched the star stand still. Where the epochs fall in time therefore decides what the fit can recover. We compared four designs at a fixed budget of 131 epochs, all restricted to Sgr A*'s observing season from Paranal (∼March–September; GRAVITY Collaboration et al. 2026).

Design A (heuristic) places about 90 per cent of the epochs within ±20 d of each periapsis and spreads the rest thinly.

Design B (D-optimal) lets the statistics choose. The Fisher information predicts how much each candidate epoch shrinks the joint uncertainty on the seven fitted parameters; a D-optimal design minimizes the volume of the joint uncertainty ellipsoid, evaluated at one assumed-true parameter vector and built greedily from a daily, season-filtered candidate grid.

Design C (pseudo-Bayesian D-optimal) averages the Fisher information over 300 draws from the published element uncertainties (K. Chaloner & I. Verdinelli 1995), discounting epochs that are informative only if the orbit is already known exactly.

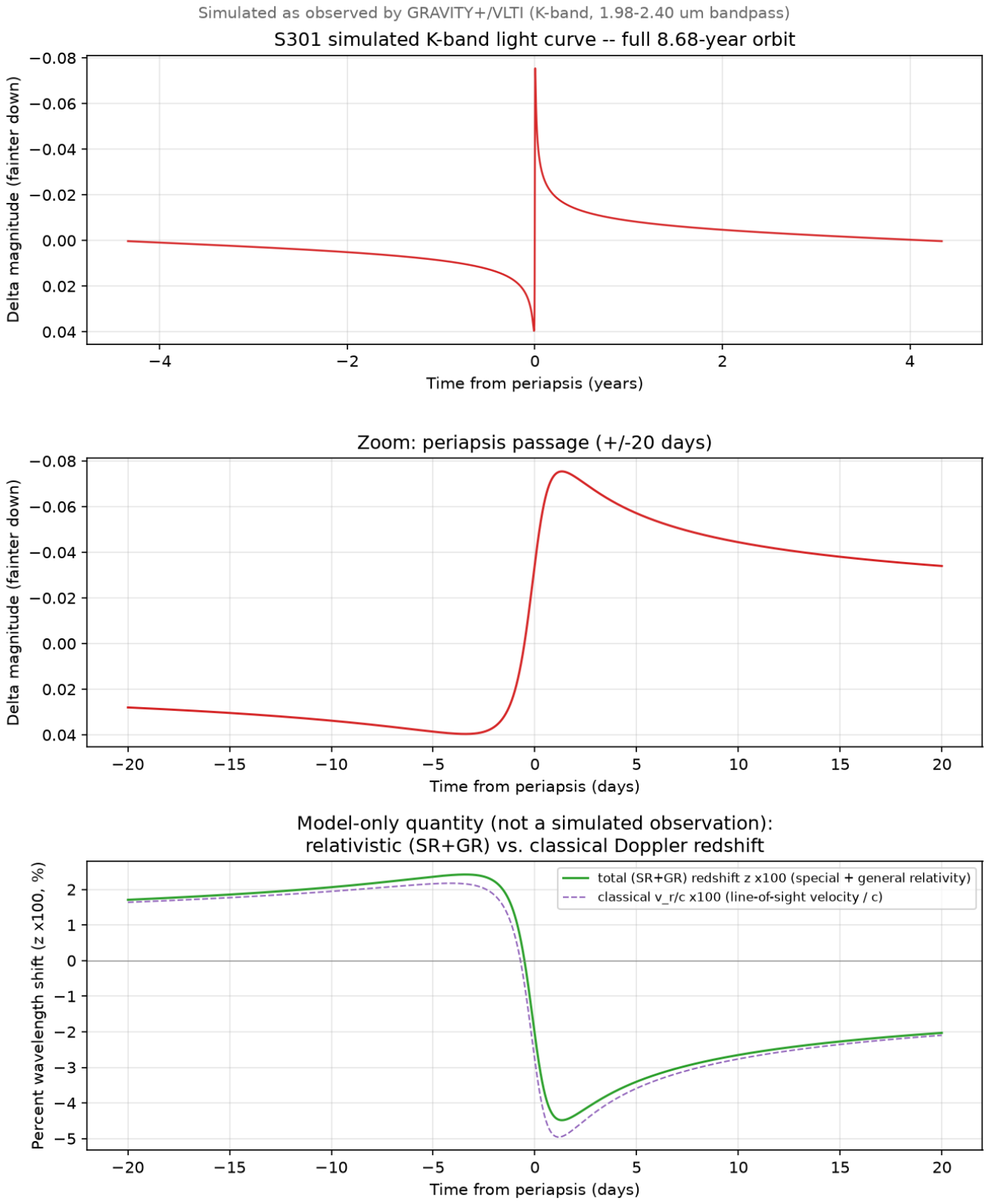


Figure 3. Simulated relativistic K-band light curve of S301 over one orbit (top), the periapsis passage (middle), and the relativistic and classical spectral shifts near periapsis (bottom). The lower panels show a generic passage in time from periapsis; precession changes only the orbit's orientation on the sky, so the shape is the same for either real passage. The bottom panel is a model-internal quantity (SR: special-relativistic Doppler shift; GR: gravitational redshift) that explains the magnitude curves above it; no instrument in this campaign measures a spectral shift.

Table 2. Cadence designs at $N = 131$: bootstrap 1σ uncertainties (1000 resamples) for Designs A–D, the Cramér–Rao bound for the adopted Design D, and the published uncertainties. Epochs within ±20 d of a periapsis: A 118, B 27, C 28, D 50.

Parameter	A	B	C	D	D (CRB)	Published
P (yr)	0.0003	0.0004	0.0005	0.0004	0.0004	0.1100
e	0.0001	0.0001	0.0001	0.0001	0.0001	0.0010
i (deg)	0.0981	0.0567	0.0677	0.0728	0.0671	1.1000
Ω (deg)	0.2511	0.1351	0.1588	0.1692	0.1611	3.5000
ω (deg)	0.1924	0.0766	0.0895	0.0953	0.0914	2.2000
t_0 (yr)	0.00054	0.00078	0.00078	0.00066	0.00072	0.0100
$\dot{\omega}$ (deg yr^{-1})	0.00890	0.00220	0.00225	0.00231	0.00216	···

Design D (constrained) runs Design C's search after reserving 0.40 of the budget for the ±20 d windows around both periapses. The season filter removes roughly half of passage 1's window, so 0.382 of the budget survives there.

Table 2 gives each design's bootstrap precision, the analytic Cramér–Rao bound (CRB) at the truth, and the number of epochs near periapsis. Design A is the outlier: its $\dot{\omega}$ precision is 3.85× worse than Design D's, because a periapsis-heavy budget leaves too little angular diversity to fix the orbit's orientation. Designs B, C and D agree on $\dot{\omega}$ within bootstrap noise, and all place many epochs near periapsis unprompted (27, 28 and 50 of 131). We adopt Design D because its floor guarantees periapsis coverage whatever the search does, and complete periapsis coverage is what pins the period and periapsis epoch (Section 4.2). For Design D the CRB and the bootstrap agree (CRB/bootstrap = 0.93 for $\dot{\omega}$), the best available evidence that the bootstrap uncertainties are sound.

The floor has a history. An earlier version of the code, with no season filter and an unsafeguarded Kepler solver, produced a Design B that piled epochs at the ends of the campaign, skipped periapsis and failed its own bootstrap. That pitfall of an unfiltered candidate grid on a nonlinear periodic problem does not recur in the current code; we keep the floor as insurance. T. Piran et al. (2026)'s multi-star design faces a different constraint, synchronizing observations across several stars to subtract Newtonian confusion; with one star and one observable, periapsis coverage is what matters.

3.4. *Solar conjunction*

The season filter exposes a hard limit. Sgr A*'s solar elongation is below 45°, and the field unobservable from any facility, from 2031-11-06 to 2032-02-01, reaching 5.6° on 2031-12-19. The first periapsis (2031-10-22, elongation 59°) falls inside this window. The last observable epoch before it is 22 d before periapsis and the first after it is 131 d after: a 153-d blackout covering the periapsis and the start of the recovery. No facility escapes it. The elongation stays below 85°, the sunshield limit of JWST (James Webb Space Telescope), from 2031-09-26 to 2032-03-12; the ELT site is 20 km from Paranal and shares its sky; and the limit is geometric, so a spectrograph sees no better than an imager. The second periapsis, at elongation 170°, is comfortably in season. The Sun, not the black hole, decides which passage we see. With $P = 8.68$ yr each periapsis arrives 0.68 yr later in the calendar than the last, and in 2031 the calendar puts it behind the Sun; in 2040 it does not. Section 4.2 quantifies the cost.

3.5. *Orbit fit*

We fit seven parameters to the astrometry: the six Keplerian elements and the apsidal precession rate $\dot{\omega}$, applied as $\omega_{\rm eff}(t) = \omega + \dot{\omega}\,(t - t_{\rm ref})$. The model evaluates the orbit at each epoch's emission time, solved from the arrival time by Newton iteration, so the Rømer delay is in both the truth and the model, as in the discovery analysis. The fit is Levenberg–Marquardt least squares started from the published solution, perturbed by 3 per cent in period, 0.03 in eccentricity, 10° in inclination, 15° in node, 12° in argument of periapsis, 0.05 yr in periapsis epoch and 0.1 deg yr^{-1} in precession rate, to stand in for the uncertainty of a preliminary orbit. The semi-major axis follows from $(P, GM_\bullet)$ by Kepler's third law with $M_\bullet$ and R_0 fixed at their published values; Section 5.3 tests the cost of freeing them. Uncertainties come from 1000 bootstrap resamples of the epochs, cross-checked against the CRB (Table 2) and against

Table 3. Final result: Design D cadence, astrometry only, 207 μas per epoch, 1000 bootstrap resamples.

Parameter	Truth	Fitted	σ	Published σ
P (yr)	8.6800	8.6800	0.0004	0.1100
e	0.9832	0.9833	0.0001	0.0010
i (deg)	124.0900	124.1368	0.0728	1.1000
Ω (deg)	73.8000	73.6504	0.1692	3.5000
ω (deg)	293.4000	293.3197	0.0953	2.2000
t_0 (yr)	2023.1260	2023.1261	0.0007	0.0100
$\dot{\omega}$ (deg yr^{-1})	0.2307	0.2324	0.0023	···

30 independent noise realizations (Section 4.2). We compare the recovered $\dot{\omega}$ with the Schwarzschild prediction,

$$\dot{\omega} = \frac{6\pi\, GM_\bullet}{c^2\, a\,(1-e^2)\, P}, \tag{1}$$

the Mercury-perihelion formula applied to S301. We inject no Lense–Thirring term, so i and Ω stay fixed: the spin of Sgr A* is unmeasured, and assuming a value would substitute a guess for a measurement.

4. RESULTS

4.1. *The recovered precession*

The adopted campaign has 131 epochs between 2028.50 and 2041.69. Of these, 50 (38.2 per cent) fall within ±20 d of a periapsis, above Design D's realized floor, because the Fisher search placed some of its flexible budget near periapsis too. The remaining 81 epochs (61.8 per cent) form a few tight clusters separated by multi-year gaps (Fig. 4), in 2033 and 2037–2038 rather than immediately after passage 1; Section 4.2 tests whether that is wasteful.

Table 3 gives the fit. Every parameter is recovered within 0.97σ of the truth, and the fitted sky-plane track (Fig. 4, top) shows the periapsis direction rotated between passages, the geometric signature of precession. Several uncertainties are far tighter than the published ones (0.1692° against 3.5° on Ω, for example) because of three idealizations: the same precision at every epoch, two full passages the 2026 data set does not yet have, and fixed $M_\bullet$ and R_0, which pin the orbit's angular scale (Section 5.3). The comparison is not a claim to outperform the discovery data.

If GRAVITY+ reaches its forecast precision (100 μas instead of 207 μas), the same campaign gives $\dot{\omega} = 0.2315 \pm 0.0012$ deg yr^{-1}, a 198σ detection instead of 96σ, with the orientation angles improved in proportion ($\sigma_\Omega = 0.0812°$). Precision scales linearly with the per-epoch error, so Table 3 can be rescaled to whatever GRAVITY+ delivers in 2031. These are white-noise figures; Section 5 is where they meet reality.

The fitted $\dot{\omega}$ is a free parameter, determined by geometry with no reference to equation (1). As a check, for each bootstrap resample we evaluate that formula on the resample's own fitted (P, e), still blind to any published or injected value: the prediction is 0.2318 ± 0.0011 deg yr^{-1}, and its per-resample difference from the fitted $\dot{\omega}$ is 0.0005 ± 0.0029 deg yr^{-1}, 0.16σ from zero (the difference is taken per resample because P, e and $\dot{\omega}$ come from the same correlated resamples). Geometry and theory agree without either referring to the other's answer.

4.2. *Testing the cadence*

Two features of the adopted cadence invite suspicion, the truncated first passage and the late clusters, and we tested both. First, does the conjunction gap cost anything? We built a design identical to Design D except that passage 1's cluster is a complete, symmetric ±20 d window centred on periapsis, and bootstrapped both. Completing the passage tightens P and t_0 to 0.60× and 0.48× their Design D values and the orientation angles by a few per cent (0.95, 0.95 and 0.97×), while $\dot{\omega}$ stays at 1.04×, unchanged within bootstrap noise. The gap costs the period and periapsis epoch and nothing measurable on $\dot{\omega}$, whose precision comes from comparing the orbit's orientation across the ∼8.7 yr between the two periapsis-anchored clusters.

Second, given that the gap is unavoidable, where should the freed budget go? Moving the 38 flexible epochs into the earliest window the season permits (2032.16–2032.75, starting 131 d after periapsis) makes $\dot{\omega}$'s precision 1.52×

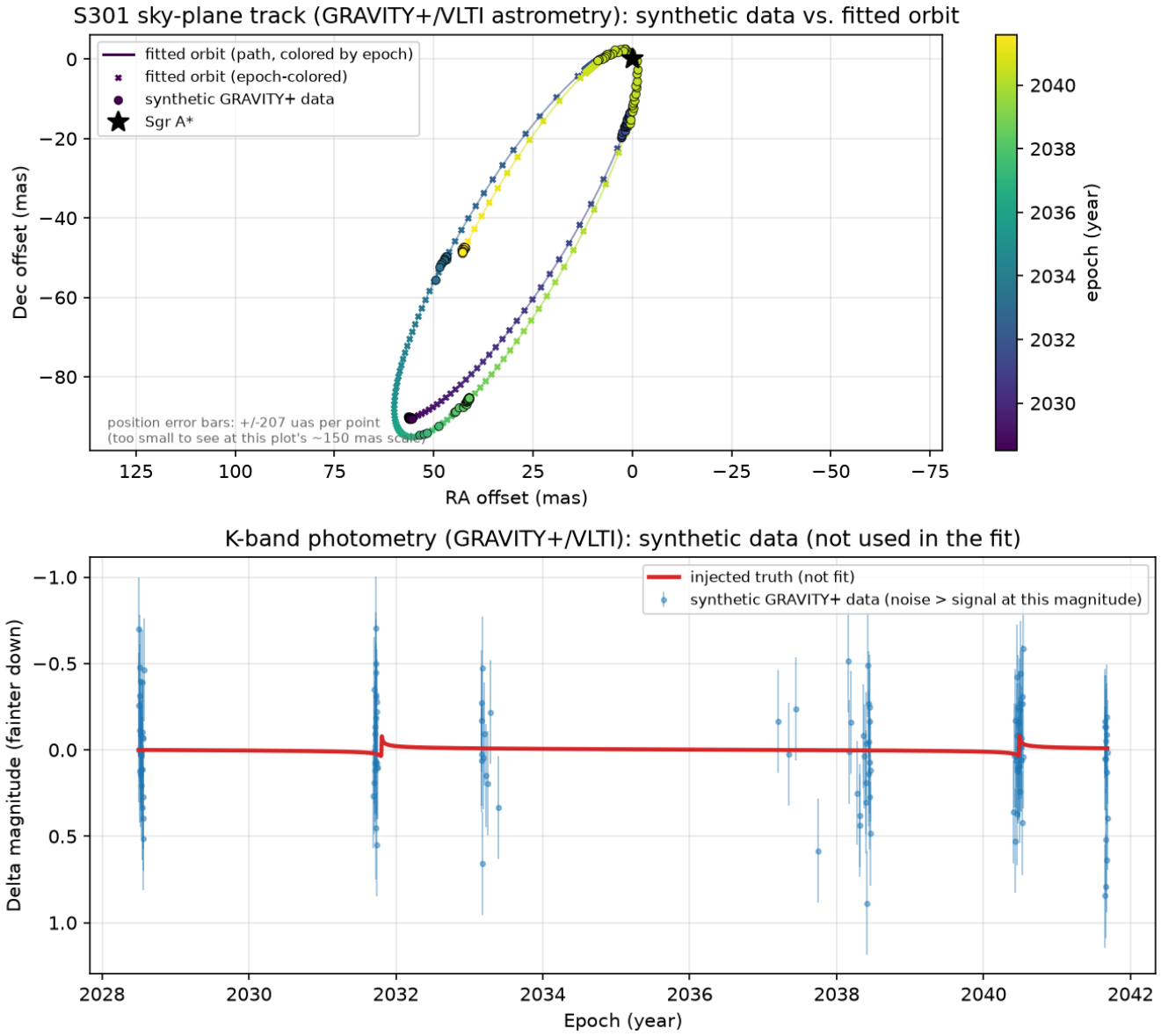


Figure 4. Top: fitted sky-plane track against the synthetic GRAVITY+ astrometry, coloured by epoch, showing the two passages of the precessing orbit. Bottom: simulated K-band photometry against the injected truth curve (not used in the fit). An animated version (`s301_precession.mp4`, an ancillary file of this submission, also at https://github.com/jcatanza/s301-schwarzschild-precession/blob/master/output/s301_precession.mp4) extrapolates the same precession rate over 20 orbits (∼174 yr) to make the rosette visible; the two orbits analysed here are highlighted and the extrapolation is labelled on screen throughout.

Design D's, 52 per cent worse, and every placement tried between 2032 and 2037 followed the same pattern (best at 2037.5, 0.00239 deg yr^{-1}; worst at 2034.5, 0.00479). Watching S301 for another month after periapsis means watching it slow down along a path the dense cluster has already traced. The fit learns more from seeing the star somewhere new: epochs at a different orbital phase break the degeneracies (period against periapsis epoch, node against inclination) that would otherwise blur the comparison between the two anchors. Finally, 30 independent noise realizations corroborate the bootstrap: the realization-to-realization scatter of $\dot{\omega}$ is 0.78× the bootstrap σ.

4.3. *Photometry as an independent check*

Every astrometric epoch also yields a K-band brightness, which we do not fit. Astrometry alone fixes the orbital state at every epoch, and photometry is a function of that state with no free parameter of its own; at S301's photometric uncertainty (0.30 mag against a 0.112 mag modulation) it adds nothing, leaving $\dot{\omega}$'s bootstrap precision at 1.03× the astrometry-only value and every other parameter within bootstrap noise. Photometry serves as a validation instead. We take the astrometry-only solution of Table 3, predict the brightness curve from Section 3.1, and compare it with the simulated photometry the fit never saw: $\chi^2 = 137.5$ for 131 epochs (reduced $\chi^2 = 1.05$, 0.40σ from expectation). An orbit derived from positions alone correctly predicts a brightness it was never fitted to. The check must be statistical: in the bottom panel of Fig. 4 the per-epoch scatter exceeds the whole signal, and the beaming spike cannot be seen by eye.

5. SYSTEMATIC ERROR BUDGET

The uncertainties in Table 3 assume white noise: every epoch an independent Gaussian draw, every error averaging down as $\sqrt{N}$. This section asks what else can go wrong, in three tiers. Terms we can model go into the fit, exactly (the Rømer delay) or as nuisance parameters that are marginalized (the Sgr A* zero point, the mass and distance). Terms we can bound but not fit are injected into the synthetic data, refitted with the naive model, and carried as biases (extended mass, lensing, confusion) or as excess scatter (systematics correlated across epochs). For whatever remains unnamed we apply the excess scatter that real GRAVITY orbit fits have shown over their formal errors. Every amplitude is anchored to a published figure. Table 4 collects the terms and Section 5.9 combines them.

5.1. *Rømer delay*

Every position we record is old news, and not all of it equally old. Light from the far side of the orbit reaches us up to $2a/c \approx 7.9$ d later than light from the near side, and because the star moves along the line of sight, the delay changes around the orbit. That swing is comparable to the ±20 d periapsis windows; over the campaign the correction ranges from -0.06 to 5.60 d (RMS 3.73 d). The pipeline includes it (Section 3.5), so Table 3 is unaffected. Ignoring it would matter: a light-time-unaware model fitted to light-time-correct, noise-free data biases $\dot{\omega}$ by -0.0023 deg yr^{-1} (1.00 per cent of the signal, 1.0σ) and i, Ω and ω by 0.53°, -1.40° and -0.75° (7–8σ). The correction is a deterministic function of the orbit being fitted and adds no free parameter: the aware model reproduces the noise-free data to machine precision (1.2×10^{-13} deg yr^{-1} residual), and on noisy data its bootstrap precision is 1.00× the unaware model's. The correction is exact in bias and free in precision.

5.2. *The Sgr A* reference zero point*

The limiting systematic of the S2 precession measurement was the tie between the NACO adaptive-optics imaging frame and the radio frame (GRAVITY Collaboration et al. 2020). It does not apply here: GRAVITY's positions "are directly referring to Sgr A*, since it is visible in each exposure" (GRAVITY Collaboration et al. 2022), and the S301 discovery fit allows "no coordinate system offsets" for the same reason (GRAVITY Collaboration et al. 2026). An earlier draft of this paper injected the NACO priors of P. M. Plewa et al. (2015); that analysis has been removed as inapplicable.

Two effects remain. The first is a constant offset between the near-infrared photocentre of Sgr A* (the accretion flow and its flares) and the mass centroid the orbit is bound to. The emitting region is a few R_S across ($1\,R_S = 10$ μas; the EHT ring is 51.8 μas, Event Horizon Telescope Collaboration 2022), so we adopt a 50 μas prior with 100 μas as the conservative case. Two extra parameters (x_0, y_0) absorb an offset common to every epoch exactly; the seven-parameter model, which places the focus at the origin, is biased by it. The second is per-epoch jitter of the photocentre. GRAVITY's quoted 65–100 μas S2-to-Sgr A* precision (GRAVITY Collaboration et al. 2020) already includes this, so it sits largely inside the campaign's 207 μas floor; we add 50 and 100 μas in quadrature as a sensitivity test.

Injecting a one-sigma offset (35.4 μas per coordinate) and refitting the seven-parameter model biases $\dot{\omega}$ by -0.0005 deg yr^{-1} (0.2σ, 0.22 per cent of the signal); the conservative offset gives 0.4σ (0.44 per cent). Fitting (x_0, y_0) jointly with the prior recovers the offset (33.4, 26.6 μas against 35.4), leaves a 0.026 per cent residual in $\dot{\omega}$, and costs 1.04× in precision ($\sigma = 0.0024$ deg yr^{-1}). Jitter of 50 or 100 μas changes the precision by 1.03× and 1.03×. The reference systematic is below 1σ if ignored and nearly free once modelled.

5.3. *Mass and distance priors*

Table 3 holds $M_\bullet$ and R_0 fixed, which pins the orbit's angular scale (a/R_0 with $a^3 = GM_\bullet P^2/4\pi^2$) by the period alone. Real S-star fits float both. Astrometry alone constrains only $GM_\bullet/R_0^3$, the mass–distance degeneracy that radial velocities normally break, so we refit with $M_\bullet$ and R_0 free under Gaussian priors from GRAVITY Collaboration et al. (2022) (statistical and systematic combined: 0.97 and 0.38 per cent). The precision on P is nearly unchanged (0.97×), because the period comes from timing; e, i and ω widen to 1.36, 1.56 and 1.51× their fixed-scale values, and $\dot\omega$ to 1.04× (0.2320 ± 0.0024 deg yr^{-1}, a shift of 0.17σ). The fixed scale was over-constraining the orbit's shape and orientation, as expected, while the precession rate is only modestly affected. The recovered ratios ($M/M_{\rm pub} = 1.0018 \pm 0.0031$, $R_0/R_{0,\rm pub} = 0.9992 \pm 0.0014$) return the priors, as astrometry-only data must.

5.4. *Extended mass*

Undetected extended mass inside S301's orbit shifts the periapsis backwards and biases the apsidal precession low (G. F. Rubilar & A. Eckart 2001). We reproduce their post-Newtonian treatment: a point mass plus a Plummer distribution ($\alpha = 5$), integrated through two periapsis passages with their Eq. A.7 equation of motion, measuring the rotation of the eccentricity vector between them. With no extended mass the integrator matches equation (1) to 1.19 per cent, the second-post-Newtonian scale $(v/c)^2 \approx 0.7$ per cent at S301's periapsis speed; the bias is a difference between two runs, so this residual largely cancels. For the mass and core radius we take GRAVITY Collaboration et al. (2022)'s multi-star constraint: Plummer scale length $0.3'' \approx 12.04$ mpc (milliparsec) with enclosed mass $\lesssim$3000 $M_\odot$ (1σ) or $\lesssim$7500 $M_\odot$ (3σ). The bias on $\dot\omega$ is -0.00013 deg yr^{-1} at the 1σ bound (0.056 per cent of the signal, 0.06σ of the random precision) and -0.00032 deg yr^{-1} at 3σ (0.137 per cent, 0.14σ). S301's periapsis (0.056 mpc) lies so deep inside any plausible core that little of the bounded mass is enclosed. T. Piran et al. (2026) model a different quantity, the nodal precession from a flattened disc competing with the spin signal, but likewise find S301 about ten times better protected than S2.

A smooth profile misses one risk. At S301's radius the discovery paper finds heavy-tailed perturbations from a few massive bodies, so that orbital evolution "would be dominated by large, non-local jumps" (GRAVITY Collaboration et al. 2026, Methods), with relaxation and collision time-scales $\gtrsim 10^7$ yr against a precession time-scale of 1.6×10^3 yr. Over two passages the expected stochastic change in ω is orders of magnitude below the signal, but a single close encounter is a rare, unbounded event that no smooth budget captures. We flag it without modelling it.

5.5. *Source confusion*

Any neighbouring star within about one diffraction-limited beam blends into the same single-mode-fibre output and shifts the recovered photocentre by its flux-weighted contribution. We bound this with a Monte Carlo of 20000 field realizations around S301 ($m_K = 19.3$), drawing neighbour counts from GRAVITY Collaboration et al. (2022)'s surface density (>100 stars per square arcsecond to $K < 17$, extended to $K = 21$ with the number–magnitude slope quoted by G. F. Rubilar & A. Eckart from Jaroszyński 1999) and weighting by the 60 mas single-telescope beam of GRAVITY Collaboration et al. (2020). As a single-aperture photocentre the median bias would be 9.7 mas per epoch, far above GRAVITY's demonstrated precision, so that is the wrong quantity. GRAVITY Collaboration et al. (2022) state that interferometric astrometry is "much less affected (by a factor of several hundred)"; taking that as 200–500×, the 90th-percentile confusion floor is 37–92 μas, comparable to or below the photon-noise floor. Because neighbours move on their own orbits, confusion is per-epoch scatter, not a coherent drift; propagated as such at the conservative end, it changes $\dot\omega$'s precision by a factor 1.00 and shifts the best fit by -0.00000 deg yr^{-1}.

5.6. *Gravitational lensing*

Sgr A* bends S301's light when the star is on the far side of the black hole. For a point lens the primary image of a source at separation θ_s appears at $\theta_+ = [\theta_s + (\theta_s^2 + 4\theta_E^2)^{1/2}]/2$, with $\theta_E^2 = (4GM_\bullet/c^2)\, D_{LS}/(D_L D_S)$ and D_{LS} the star's depth behind the lens. We apply the exact shift epoch by epoch to the noise-free astrometry and refit. Only 4 of the 131 epochs are on the far side; the largest shift is 22.2 μas ($\theta_E = 163$ μas) at the far-side periapsis of passage 2, 1 epoch exceeds 10 μas, and the RMS shift is 2.05 μas. The lens-unaware fit is biased by -0.00000 deg yr^{-1} in $\dot\omega$ (0.00σ) and by at most 0.01σ in any orientation angle. Lensing is real, computable and negligible for this campaign.

5.7. *Correlated systematics*

White noise is forgiving: average enough of it and it cancels. Errors that move together across a run or a season do not cancel, and the epoch bootstrap, which shuffles epochs as if each were a fresh draw, cannot see them at all.

We model two, each at a published amplitude, on top of the white noise. The first is a per-run calibration offset. GRAVITY Collaboration et al. (2022) find that their calibration "adds a systematic uncertainty of 60 μas, divided by the square root of the number of available calibrations"; we take one calibration per observing run, defined as epochs separated by less than 30 d, so the 131 epochs form 12 runs and every epoch in a run shares one $N(0, 60\,\mu\text{as})$ offset per coordinate. A dense periapsis cluster is then a single run, the conservative reading. The second is a slowly wandering reference. The near-infrared photocentre of Sgr A* is the accretion flow, and the centre of its flare orbits agrees with the mass centroid only "to within the $\pm$50 μas uncertainties" (GRAVITY Collaboration et al. 2018b); slowly moving confusing neighbours behave the same way. We model this as a red-noise (Ornstein–Uhlenbeck) process with standard deviation 50 μas per coordinate and a correlation time of 0.5 yr, one observing season. The fit is told only the white errors, as a real analysis would be.

Because the bootstrap is blind to these terms, uncertainties here come from 40 independent realizations of the whole campaign. With white noise alone the realization scatter of $\dot{\omega}$ is 0.00167 deg yr^{-1} (0.72$\times$ the Table 3 bootstrap). With both correlated terms it grows to 0.00349 deg yr^{-1}, 2.09$\times$ the white value, with a mean bias of -0.00027 deg yr^{-1} (0.08σ). Scanned separately, common-mode offsets of 30, 60 and 100 μas inflate the precision by 1.13, 1.66 and 2.55$\times$, and red noise of 50 and 100 μas by 1.53 and 2.66$\times$. The epoch bootstrap applied to one correlated realization, the estimator a real analysis would reach for first, reports only 0.71$\times$ the true scatter. Section 5.9 carries the excess into the bottom line.

5.8. *Spin: the known unknown term*

Every term above can be bounded. The spin of Sgr A* cannot: for a spin aligned with S301's orbital angular momentum, frame dragging produces a purely apsidal shift (`orbit.lense_thirring_apsidal_rate`, validated against the discovery paper's 0.114 deg per orbit at $\chi = 1$) that enters $\dot{\omega}$ with the same sign and the same time dependence as the Schwarzschild term, and with no upper limit to inject. Using the discovery paper's fiducial case ($\chi = 1$, aligned) and an independent outflow-method measurement ($\chi = 0.90 \pm 0.06$; R. A. Daly & et al. 2024), the possible contamination is 0.01192–0.01325 deg yr^{-1}: 5.17–5.74 per cent of the Schwarzschild signal, or 5.2–5.7σ of the random precision (5.0–5.5σ of the reference-marginalized precision), larger than any single term in Table 4. A measured $\dot{\omega}$ from S301 alone cannot separate pure Schwarzschild precession from Schwarzschild plus aligned frame dragging; T. Piran et al. (2026)'s multi-star method exists to break that degeneracy.

5.9. *Combined budget and headline precision*

Multiplying the separate precision costs of the reference offset (1.04$\times$) and the mass–distance priors (1.04$\times$) would assume they are independent. Instead we fit all four nuisance parameters jointly with the orbit (11 parameters) and bootstrap: the marginalized precision is 0.0024 deg yr^{-1}, 1.04$\times$ the random-only value. Adding the bounded biases in quadrature (extended mass at the 3σ bound, lensing, and the confusion-inflated variance) gives the white-noise bottom line, $\pm$0.0024 deg yr^{-1}. Carrying the correlated-systematics excess of Section 5.7 in the same way gives $\pm$0.0050 deg yr^{-1}, 2.17 per cent of the signal; correlated systematics supply 77 per cent of that variance.

One allowance remains, for what this budget has not named. Real GRAVITY orbit fits scatter more than their formal errors predict: the S2 precession fit has reduced $\chi^2 = 1.5$ (GRAVITY Collaboration et al. 2020) and the four-star fit of GRAVITY Collaboration et al. (2022) has $\chi^2_r = 2.17$. Precision scales linearly with the per-epoch error, so inflating by $\sqrt{\chi^2_r} = 1.22$–1.47 gives, at the conservative end, $\pm$0.0074 deg yr^{-1}: 3.20 per cent of the signal, a 31σ detection. That is the number we would defend in front of a time-allocation committee, and it is the headline of this paper. The S301 discovery fit itself shows no such excess ($\chi^2 = 26$ for 34 degrees of freedom at the 207 μas floor; GRAVITY Collaboration et al. 2026), so the inflation is an envelope rather than a prediction. More telescope time still helps the white-noise term (doubling the budget to 262 epochs improves it by 29 per cent, consistent with $\sqrt{N}$), but the correlated terms average down only with the number of runs and seasons, which argues for spreading a fixed budget over more independent visits rather than longer ones. Nothing in the budget bounds the spin term, which at published spin values is 1.6–1.8σ of the headline precision. That term makes the result a *spin-agnostic* test (Section 6.3); closing the gap is T. Piran et al. (2026)'s multi-star programme, beyond this pipeline.

6. DISCUSSION

6.1. *Why GRAVITY+ and not another instrument*

No operating alternative matches GRAVITY+ for this measurement, and the comparison comes down to resolution. The 2.4 m aperture of the Roman Space Telescope has a diffraction limit of order 223 mas, larger than S301's entire

Table 4. Systematic error budget for $\dot{\omega}$ (deg yr^{-1}).

Term	Type	Effect	Status
Photon noise, 207 μas	Random	$\pm$0.0023	Baseline (Table 3)
Photon noise, 100 μas	Random	$\pm$0.0012	If GRAVITY+ meets its forecast
Rømer delay, if ignored	Geometric	-0.0023 (1.0σ)	Not incurred: in the model
Reference offset, ignored	Instrumental	-0.0005 to -0.0010 (0.2–0.4σ)	Sub-σ even if ignored
Reference offset, fit jointly	Instrumental	residual 0.026%; precision $\times$1.04	Marginalized
Reference jitter, 100 μas	Instrumental	precision $\times$1.03	Inside the floor
$M_\bullet$, R_0 free (priors)	Model	precision $\times$1.04	Marginalized
All nuisances jointly	Model	$\pm$0.0024 ($\times$1.04)	Marginalized
Extended mass (1–3σ bound)	Astrophysical	-0.00013 to -0.00032 (0.06–0.14σ)	Bounded, in quadrature
Source confusion	Instrumental	precision $\times$1.00	Subdominant, in quadrature
Lensing	Geometric	-0.00000 (0.00σ)	Negligible, in quadrature
White-noise bottom line		$\pm$0.0024 ($\times$1.05)	Nuisances + bounded biases
Correlated systematics, 60 + 50 μas	Instrumental	precision $\times$2.09; bias -0.00027	Realization scatter, in quadrature
Bottom line		$\pm$0.0050 ($\times$2.17, 2.17% of signal)	With correlated systematics
Empirical inflation, $\sqrt{\chi^2_r} = 1.47$	Unknown unknowns	$\pm$0.0074 (3.20% of signal)	**Headline: 31σ**
Sgr A* spin, $\chi = 0.9$–1	Degeneracy	+0.01192 to +0.01325 (1.6–1.8σ of headline)	Unbounded; out of scope

NOTE—Sections 5.1–5.8 give each term's sources and method, in the same order. Ignored Rømer delay would also bias i, Ω, ω by 7–8σ. The white-noise bottom line is the joint 11-parameter bootstrap (orbit, reference offset, $M_\bullet$, R_0) combined in quadrature with the 3σ extended-mass bias, the lensing bias and the confusion-inflated variance; the bottom line adds the correlated-systematics excess of Section 5.7; the headline multiplies it by the $\sqrt{\chi^2_r}$ of the four-star GRAVITY fit of GRAVITY Collaboration et al. (2022).

83 mas orbit. The 6.5 m aperture of JWST reaches ∼85 mas, comparable to the orbit itself. The ∼130 m baseline of GRAVITY+ gives a diffraction limit of order 3 mas and astrometry at the 100–200 μas level. Only the last resolves the orbit's shape, let alone its precession.

6.2. *Prospects for a velocity channel*

A simulated radial-velocity channel at ERIS precision (1621 km s^{-1}) changes $\dot{\omega}$'s precision by 4.5 per cent, within bootstrap noise, so no ERIS time is requested. Roman's slitless spectroscopy ($R \approx 461$–890) fails regardless of integration time because its point spread function is not adaptive-optics-corrected in this crowded field. JWST/NIRSpec has the resolution (A. J. Shajib et al. 2025) but reported operational problems in the Galactic Centre (saturation on bright cluster members, micro-shutter cross-talk, guide-star misidentification; F. Yusef-Zadeh et al. 2025) and no published precision for a source this faint.

The ELT is the real prospect. The discovery team plans MICADO spectroscopy, which they expect "will have no problem" measuring S301's velocity, and their spin forecast assumes 1 km s^{-1} (GRAVITY Collaboration et al. 2026). Our estimate is more cautious: the background-limited scaling of Section 3.2, applied to HARMONI's 39 m aperture and $R \approx 17{,}385$ K-band modes (N. Thatte & et al. 2024), predicts $\sigma_{\rm RV} \approx 20.6$ km s^{-1} (signal-to-noise ratio of order 741), a factor ∼20 short of the MICADO figure. Neither number rests on an achieved observation, and the MICADO figure is stated without a sensitivity calculation. Either would be useful: with the HARMONI figure a velocity channel tightens P, e and t_0 by tens of per cent and the orientation angles by a quarter to a third, but changes $\dot{\omega}$'s precision by -4.1 per cent, within bootstrap noise, because two-passage astrometry already constrains the precession rate tightly. ESO's schedule puts ELT first light in 2029 and HARMONI science operations from late 2030 (ESO 2026), so passage 1 (October 2031) is a stretch goal for any ELT instrument and passage 2 (2040.5) is comfortable. A velocity channel by passage 2 is worth planning for, chiefly to break the mass–distance degeneracy of Section 5.3.

6.3. *Scope of the test*

The quantity measured, $\dot{\omega}$ in equation (1), is the Schwarzschild (first-post-Newtonian, monopole-mass) apsidal precession. Any sufficiently compact central mass produces it, spinning or not; the Sun's spin plays no role in Mercury's perihelion advance. Frame dragging from the spin of Sgr A* adds nodal precession and, for an aligned spin, further apsidal precession (Section 5.8). We call the measurement *spin-agnostic* in a precise sense: it needs no knowledge of the spin to be a valid test, but the spin can still change the number reported, by up to 5.74 per cent if non-zero and aligned. Measuring the spin is outside this paper's scope, as, for now, is real data: like T. Piran et al. (2026), it is a forecast. What it completes for the eventual spin measurement is a validated light-curve and astrometric pipeline, a tested cadence, and the budget of Table 4.

6.4. *Lessons*

Fisher-information design assumes that the model is close to linear around the assumed parameters; averaging the information over a prior (K. Chaloner & I. Verdinelli 1995), as in Designs C and D, is the standard remedy and costs nothing here. The broader lesson of this project concerns verification. The most consequential errors we found were a Kepler solver that diverged silently for ordinary mean anomalies at $e = 0.98$ (now a bisection-safeguarded Newton iteration, verified on 1.4×10^6 random cases to $e = 0.9999$), a photometric proxy with twice the amplitude of the validated model, a candidate grid that ignored solar conjunction, and a systematic imported from a different instrument. Each was caught only by running a real nonlinear fit or re-reading a primary source. For a problem this nonlinear, the analytic estimate is a promise and the fit is the receipt; we learned not to publish the promise.

7. CONCLUSIONS

We asked whether a two-passage, astrometry-only GRAVITY+ campaign can detect the Schwarzschild precession of S301, the fastest known star in the Galaxy, and answered by building the full pipeline such a proposal would need.

1. A relativistic light-curve model reproduces S301's published orbit to better than 0.1 per cent and predicts an asymmetric beaming feature of 0.115 mag at periapsis, invisible against S301's 0.30 mag photometric uncertainty. The science rests on astrometry; the photometry is an independent check, and it passes ($\chi^2/N = 1.05$).

2. Radial velocity adds nothing at ERIS precision, so the campaign is astrometry only. Of four cadence designs at a fixed 131-epoch budget, the Fisher-optimal designs match the adopted constrained design on $\dot{\omega}$ and all beat the periapsis-heavy heuristic.

3. Solar conjunction hides the first periapsis and the first 131 d of its recovery from every facility near Earth. Completing that passage would tighten P and t_0 but not $\dot{\omega}$, and spending the freed epochs immediately after conjunction would be 52 per cent worse than the adopted late clusters.

4. With white noise at the 207 μas achieved on S301, the campaign recovers the injected rate, 0.2307 deg yr^{-1}, as 0.2324 ± 0.0023 deg yr^{-1} (± 0.0012 if GRAVITY+ reaches its forecast 100 μas); the Schwarzschild formula evaluated on the fit's own period and eccentricity agrees to 0.16σ.

5. Of the white-noise systematics none dominates: extended mass biases $\dot{\omega}$ by at most 0.14σ; the Sgr A* reference offset is below 1σ if ignored and nearly free once fitted; confusion and lensing are negligible; the Rømer delay is in the model; freeing $M_\bullet$ and R_0 costs 1.04×. The largest correction is for errors correlated across epochs, per-run calibration offsets and a wandering Sgr A* photocentre at GRAVITY's published amplitudes, which inflate the precision by 2.09×. With those, the marginalized nuisances, the bounded biases and the empirical $\sqrt{\chi^2_r}$ inflation of real GRAVITY fits, the realistic precision is 0.0074 deg yr^{-1}, a 31σ detection.

6. An aligned spin of Sgr A* at published estimates would shift the rate by 5.17–5.74 per cent with no bound to subtract. The campaign is therefore a spin-agnostic test of general relativity: valid without knowing the spin, but not a measurement of it.

This is a forecast, and the star keeps its own schedule. Its 2031 periapsis falls behind the Sun, so the first passage can be observed only on the approach and the late recovery; the 2040 periapsis falls in full view. If the campaign is carried out as designed around those constraints, the instruments already exist for an independent, spin-agnostic test of general relativity around Sgr A*, before the harder spin measurement that S301 was found to make.

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This work uses the published orbital solution of GRAVITY Collaboration et al. (2026) and engages with T. Piran et al. (2026)'s multi-star methodology for comparison without adopting it.

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DATA AVAILABILITY

All code, results files, figures and the precession animation, from which every number in this paper is generated, are available at https://github.com/jcatanza/s301-schwarzschild-precession (code under the MIT licence; manuscript, figures and animation under CC BY 4.0). No observational data were used; the orbital elements are those published by GRAVITY Collaboration et al. (2026).

REFERENCES

Chaloner, K., & Verdinelli, I. 1995, Statistical Science, 10, 273

Daly, R. A., & et al. 2024, Monthly Notices of the Royal Astronomical Society, 527, 428. https://arxiv.org/abs/2310.12108

Davies, R. I., & et al. 2023, Astronomy & Astrophysics, 674, A207. https://arxiv.org/abs/2304.02343

ESO. 2026, Telescope first light for ESO's Extremely Large Telescope now planned for March 2029,, https://www.eso.org/public/announcements/ann25001/

Event Horizon Telescope Collaboration. 2022, The Astrophysical Journal Letters, 930, L12, doi: 10.3847/2041-8213/ac6674

GRAVITY Collaboration, Abd El Dayem, K., & et al. 2026, Nature. https://arxiv.org/abs/2607.12664

GRAVITY Collaboration, Abuter, R., Amorim, A., & et al. 2018a, Astronomy & Astrophysics, 615, L15. https://arxiv.org/abs/1807.09409

GRAVITY Collaboration, Abuter, R., Amorim, A., & et al. 2018b, Astronomy & Astrophysics, 618, L10. https://arxiv.org/abs/1810.12641

GRAVITY Collaboration, Abuter, R., Amorim, A., & et al. 2020, Astronomy & Astrophysics, 636, L5. https://arxiv.org/abs/2004.07187

GRAVITY Collaboration, Abuter, R., Amorim, A., & et al. 2022, Astronomy & Astrophysics, 657, L12. https://arxiv.org/abs/2112.07478

GRAVITY Collaboration, Abuter, R., & et al. 2017, Astronomy & Astrophysics, 602, A94. https://arxiv.org/abs/1705.02345

Koposov, S. E., Boubert, D., Li, T. S., & et al. 2020, Monthly Notices of the Royal Astronomical Society, 491, 2465. https://arxiv.org/abs/1907.11725

Peißker, F., Eckart, A., Zajaček, M., Ali, B., & Parsa, M. 2020, The Astrophysical Journal, 899, 50. https://arxiv.org/abs/2008.04764

Piran, T., Amaro-Seoane, P., Aytac, B., & et al. 2026, arXiv e-prints. https://arxiv.org/abs/2607.24931

Plewa, P. M., Gillessen, S., Eisenhauer, F., & et al. 2015, Monthly Notices of the Royal Astronomical Society, 453, 3234. https://arxiv.org/abs/1509.01941

Rubilar, G. F., & Eckart, A. 2001, Astronomy & Astrophysics, 374, 95, doi: 10.1051/0004-6361:20010640

Shajib, A. J., Treu, T., Melo, A., & et al. 2025, Astronomy & Astrophysics, 702, L12. https://arxiv.org/abs/2507.03746

Thatte, N., & et al. 2024, in Proc. SPIE, Vol. 13096, 1309614, doi: 10.1117/12.3018520

Yusef-Zadeh, F., Bushouse, H., Arendt, R. G., & et al. 2025, The Astrophysical Journal Letters. https://arxiv.org/abs/2501.04096